

Grothendieck's inequality: missing facts for randomized rounding.

let $u, y \in \mathbb{R}^n$ have entries all identically distributed to $N(0,1)$ and $A \in \mathbb{R}^{n \times n}$

Note that the entries of u , respectively y , are not necessarily independent

let $\tau > 0$ & for $i \in [n]$ define:

$$x_i^{\leq \tau} = \begin{cases} u_i & \text{if } |u_i| \leq \tau \\ \tau & \text{if } u_i > \tau \\ -\tau & \text{if } u_i < -\tau \end{cases}$$

and $u^{>\tau} = u - u^{\leq \tau}$, Analogously define $y^{\leq \tau}$ & $y^{>\tau}$, we need to show:

$$\mathbb{E} \langle x^{>\tau}, Ay^{\leq \tau} \rangle + \langle u^{\leq \tau}, Ay^{>\tau} \rangle + \langle u^{>\tau}, Ay^{>\tau} \rangle \leq 5e^{-\tau^2/4} \|A\|_G$$

where $\|A\|_G = \max_{W \succ 0} \langle W, \begin{pmatrix} 0 & A \\ 0 & 0 \end{pmatrix} \rangle$
 $\text{diag}(W) = I_{2n}$

let $w_1 = \begin{pmatrix} x^{>\tau} \\ y^{\leq \tau} \end{pmatrix}$, $w_2 = \begin{pmatrix} u^{\leq \tau} \\ y^{>\tau} \end{pmatrix}$, $w_3 = \begin{pmatrix} u^{>\tau} \\ y^{>\tau} \end{pmatrix}$

and $W = \mathbb{E} w_1 w_1^T + \mathbb{E} w_2 w_2^T + \mathbb{E} w_3 w_3^T$

Part I: verify that $\mathbb{E} \langle x^{>\tau}, Ay^{\leq \tau} \rangle + \langle x^{\leq \tau}, Ay \rangle + \langle u^{>\tau}, Ay^{>\tau} \rangle \leq \|W\| \cdot \|A\|_G$

Solution:

for any vector $u, v \in \mathbb{R}^n$, $\langle u, Av \rangle = \text{tr}(Avu^T) = \langle A, uv^T \rangle_F$ where f is Frobenius. Taking expectation & linearity

$$\mathbb{E} \langle x^{>\tau}, Ay^{\leq \tau} \rangle = \langle A, \mathbb{E} [u^{>\tau} (y^{\leq \tau})^T] \rangle_F,$$

applying these to all three terms and summing:

$$\mathbb{E} \langle x^{>\tau}, Ay^{\leq \tau} \rangle + \mathbb{E} \langle u^{\leq \tau}, Ay^{>\tau} \rangle + \mathbb{E} \langle x^{>\tau}, Ay^{>\tau} \rangle = \langle A, W^{uy} \rangle_F$$

So the its bounding equals $\langle A, W^{uy} \rangle_F$

rewrite this as an innerproduct with a block matrix

$$\left\langle \begin{pmatrix} 0 & A \\ 0 & 0 \end{pmatrix}, W \right\rangle_F = \left\langle \begin{pmatrix} 0 & A \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} W^{xx} & W^{xy} \\ (W^{xy})^T & W^{yy} \end{pmatrix} \right\rangle_F = \langle A, W^{xy} \rangle_F$$

then check that W is PSD.

$$W = E W_1 W_1^T + E W_2 W_2^T + E W_3 W_3^T \rightarrow \text{sum of PSD is PSD.}$$

Then, normalize W to have diagonal:

W is PSD but its diagonal is not 1, $\|A\|_G$ requires $\text{diag}(W) = I_{2n}$,

Claim: All diagonal entries w are equal to the same value, call it w_{ii}

$$w_{ii} = E(x_i^{>\tau})^2 + E(u_i^{\leq\tau})^2 + E(u_i^{>\tau})^2$$

Since each $x_i \sim N(0,1)$ has the same distribution, this value is the same for every $i \leq n$. By symmetry of the construction, the same value appears for $i > n$.

$2n$ diagonal entries $\Leftrightarrow w_{ii}$

• define $\tilde{W} = W/w_{ii}$ then

- $\tilde{W}$ is PSD (scaling a PSD matrix by a positive scalar keeps it PSD)

- $\text{diag}(\tilde{W}) = I_{2n}$

$\tilde{W}$ is feasible for $\|A\|_G$ maximization.

$$\|A\|_G = \max_{W' \succeq 0, \text{diag}(W') = I_{2n}} \langle W', \begin{pmatrix} 0 & A \\ 0 & 0 \end{pmatrix} \rangle \geq \langle \tilde{W}, \begin{pmatrix} 0 & A \\ 0 & 0 \end{pmatrix} \rangle$$

multiply both sides by $w_{ii} > 0$

$$w_{ii} \|A\|_G \geq \langle W, \begin{pmatrix} 0 & A \\ 0 & 0 \end{pmatrix} \rangle = E \langle u^{>\tau}, A y^{\leq\tau} \rangle + E \langle u^{\leq\tau}, A y^{>\tau} \rangle + E \langle x^{>\tau}, A y^{>\tau} \rangle$$

Part II: Rescaling with α, β

$$W_1' = \begin{pmatrix} \alpha \cdot u^{>\tau} \\ \beta \cdot y^{\leq\tau} \end{pmatrix} \quad W_2' = \begin{pmatrix} \beta \cdot x^{\leq\tau} \\ \alpha \cdot y^{>\tau} \end{pmatrix} \quad W_3' = ?$$

$$W' = E W_1' W_1'^T + E W_2' W_2'^T + E W_3' W_3'^T$$

$$E W_1' W_1'^T = \begin{pmatrix} \alpha^2 E[u^{>\tau} (u^{>\tau})^T] & \alpha\beta E[u^{>\tau} (y^{\leq\tau})^T] \\ \alpha\beta E[y^{\leq\tau} (u^{>\tau})^T] & \beta^2 E[y^{\leq\tau} (y^{\leq\tau})^T] \end{pmatrix}$$

$$E W_2' W_2'^T = \begin{pmatrix} \beta^2 E[x^{\leq\tau} (x^{\leq\tau})^T] & \alpha\beta E[x^{\leq\tau} (y^{>\tau})^T] \\ \alpha\beta E[y^{>\tau} (x^{\leq\tau})^T] & \alpha^2 E[y^{>\tau} (y^{>\tau})^T] \end{pmatrix}$$

$$E W_3' W_3'^T = \begin{pmatrix} E[u^{>\tau} (u^{>\tau})^T] & E[u^{>\tau} (y^{>\tau})^T] \\ E[y^{>\tau} (u^{>\tau})^T] & E[y^{>\tau} (y^{>\tau})^T] \end{pmatrix}$$

resulting:

$$W^{xy} = \alpha\beta E[u^{>T} | y^{<T}]^T + \alpha\beta E[u^{<T} | y^{>T}]^T + E[u^{>T} | y^{>T}]^T$$

compare with W^{yy} from part I

recall,

$$W^{yy} = E[u^{>T} | y^{<T}]^T + E[u^{<T} | y^{>T}]^T + E[u^{>T} | y^{>T}]^T$$

we need three coefficients so it will be equal:

$$\alpha\beta = 1 \quad \alpha\beta = 1 \quad 1 = 1$$

Therefore the sufficient condition is $\alpha\beta = 1$

